# Adaptive Simulation Budgets for Portfolio Decision Validation: A Paired Comparison and Feasibility Audit

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**Abstract**

We evaluate a four-checkpoint Monte Carlo validation gate as a computational allocation policy, rather than as an optimal stopping rule or a trading strategy. Candidate portfolio actions are frozen after training simulation and checked at 6, 12, 24, and 48 holdout replications. Across 66 states from one historical trajectory and 30 holdout seeds, the gate uses 9.64 replications on average and agrees with a 192-replication reference in 82.73% of seed–state observations. A retrospectively budget-matched, state-independent mixture of fixed-6 and fixed-48 gates achieves 73.46% agreement. Independent evaluation scores yield a conditional regret-proxy contrast of $-1.80 \times 10^{-6}$ engine-score units on the 965 of 1,980 observations jointly scored by all compared methods; a descriptive seed-level interval is $[-2.36, -1.25] \times 10^{-6}$. Fallback actions account for much of the missing score support. A separate paired audit shows that screening eligibility by requiring feasibility in every replication raises fallback frequency from 28.28% at budget 6 to 47.98% at budget 96, whereas a fixed-six-replication eligibility screen holds it at 28.28%. This comparison identifies a budget-dependent selection mechanism in the implemented gate. Neither the finite-budget reference nor the noisy score proxy identifies the economically optimal action, and the evidence does not establish a trading-return advantage or general optimality of the stopping policy.

**Keywords:** sequential simulation; computational budget; portfolio decision; Monte Carlo validation; regret proxy; feasibility screening.
**JEL:** C15, C61, G11.
**Code and data:** https://github.com/SequentialDecisionResearch/ADTPA-Regret (public research-code repository; historical market snapshot not redistributed).

## 1 Introduction

Portfolio policies that optimize over simulated return paths may choose among several actions whose estimated scores are close. A single evaluation budget is straightforward to audit but can spend many simulations on unambiguous decisions and too few on difficult ones. The issue is especially visible in an accumulation policy that chooses whether to buy individual equities, buy an investable benchmark, or retain cash. Candidate evaluation is expensive because the policy assesses portfolio-level constraints and future scenarios rather than a scalar return forecast.

The question examined here is narrow: can a sequential holdout gate allocate fewer replications on average than a fixed-budget gate while preserving decisions and simulation-based scores relative to a substantially larger computational reference? A second question arises from the implementation. When candidate eligibility requires feasibility in *every* simulated replication, increasing the number of replications changes the feasible action set itself. That property can make a high-budget decision a different decision problem, rather than simply a more precise estimate of the same objective.

The empirical analysis uses a fixed historical path and a frozen set of 66 portfolio states, followed by independent Monte Carlo seed replications. The principal comparison is between a four-look gate (6, 12, 24, 48) and state-independent randomized fixed-budget mixtures matched on mean simulation cost. We evaluate action agreement with 96- and 192-replication reference gates, a separate score-regret proxy, and a common-support

analysis. We then isolate budget-dependent eligibility in a small paired diagnostic. The data do not support claims of a statistically established return premium or a universally optimal sequential stopping rule.

This setting connects simulation optimization and ranking-and-selection procedures (Kim and Nelson, 2006; Fu, 2015), sequential testing (Wald, 1945; Jennison and Turnbull, 2000), and decision-focused evaluation (Elmachtoub and Grigas, 2022). It also raises the familiar difficulty of financial backtest selection (White, 2000; Gu et al., 2020). We compare the sequential gate with a state-independent randomized fixed-budget mixture matched retrospectively to its observed mean simulation cost. We also examine how requiring feasibility in every replication changes the eligible action set as the simulation budget increases. Both findings are conditional on the frozen historical state panel and the implemented simulator; they do not address investor preference learning or optimal review-time policies.

# 2 Relation to simulation selection and allocation

Ranking-and-selection methods study the allocation of simulation effort among competing systems and the probability of selecting a preferred alternative; classical procedures and their assumptions are reviewed by Kim and Nelson (2006). Optimal computing budget allocation (OCBA) targets allocation efficiency under particular simulation and asymptotic approximations (Chen et al., 2000). Sequential testing supplies another relevant design language (Wald, 1945; Jennison and Turnbull, 2000). The gate allocates additional validation batches to a frozen candidate menu rather than optimizing the number of replications assigned to individual alternatives. Its four checkpoints and Bonferroni-adjusted Student-$t$ thresholds are implementation choices, without an established asymptotic optimality or finite-sample selection guarantee. Requiring feasibility in every replication also makes the eligible menu depend on the validation budget. The principal estimand is thus agreement with a finite-budget implementation reference, accompanied by a separately scored regret proxy, rather than the probability of correct selection against a known best system. This scope is narrower than a general-purpose simulation-optimization method.

# 3 Decision setting and validation protocol

## 3.1 Frozen state and candidate menu

At a decision date $t$, the state $s_t$ contains prices observed through $t$, existing shares $q_t$, benchmark shares, cash, the dated contribution schedule, and the policy configuration. The implementation permits purchases and no-purchase actions but not sales. Its candidate menu contains action types $\mathcal{A} = \{\text{stock buy}, \text{IYW buy}, \text{cash wait}\}$, each paired with a review horizon and allocation weights. The action set is not a claim that all three actions are feasible in every state.

For each state, a training simulation of 300 scenarios constructs a candidate list and freezes one highest-scoring training-feasible candidate within each action type. Each subsequent holdout replication evaluates 250 scenarios with the same frozen candidate identities. The baseline is an IYW purchase when it is eligible. If it is not, the implementation returns an explicit infeasible fallback. Training and holdout streams use separate deterministic seed families; independent evaluation scores use a third family. These are pseudo-random simulation streams, not independent market histories.

Let $S_r(a; s)$ be the simulated score of frozen candidate $a$ at holdout replication $r$, and $F_r(a; s) \in \{0, 1\}$ its feasibility indicator. The implementation's original eligibility rule after $n$ replications is

$$\mathcal{A}_n(s) = \left\{ a \in \mathcal{A}_{\text{train}}(s) : \prod_{r=1}^{n} F_r(a; s) = 1 \right\}. \tag{1}$$

The candidates' underlying allocations are frozen within a state, but the eligible set can shrink as $n$ grows. The term budget-dependent eligibility refers to this deterministic screening dependence; it does not assert statistical bias relative to a validated population chance constraint. An IYW-ineligible state produces a fallback action, which is distinct from an IYW purchase.

## 3.2 Pairwise confidence gates

For two eligible actions $a, b$, define the paired score differences $D_r(a, b) = S_r(a; s) - S_r(b; s)$, sample mean $\bar{D}_n$ and standard error $\widehat{\text{se}}_n = s_D/\sqrt{n}$. At checkpoint $n$ a challenger can be selected if its lower confidence bound against the benchmark and each other eligible competitor exceeds the implementation's minimum advantage $\delta = 0$:

$$L_n(a, b) = \bar{D}_n(a, b) - c_n \widehat{\text{se}}_n(a, b) > \delta. \tag{2}$$

Only a unique challenger satisfying all comparisons is accepted. At an intermediate checkpoint, the procedure may stop with the IYW default when all eligible challengers have upper bounds below $-\delta$, or when no challenger remains. Otherwise it continues until the last checkpoint, where it defaults if no challenger passes. An ineligible IYW candidate instead triggers fallback. Full decisions include the action, horizon, allocations, and feasibility flag; agreement is stricter than matching action type alone.

For the four-look method, the source code uses a two-sided Student-$t$ Bonferroni critical value

$$c_n = t_{n-1}^{-1}\Big(1 - \frac{\alpha}{2KM}\Big), \qquad \alpha = 0.05,\ K = 4,\ M = 3, \quad n \in \{6, 12, 24, 48\}. \tag{3}$$

Here $M$ is a conservative count of pairwise action contrasts. The fixed-budget comparators use their corresponding single-look adjustment, while the original 6-to-12 adaptive procedure uses two looks. Consequently budget changes do not isolate precision alone: the critical value and, under (1), the eligible set may also change. Equation (3) is a *working approximation*, not a finite-sample anytime-valid confidence sequence. A formal familywise guarantee would require valid marginal tests under the actual score-generating law and the declared family of comparisons; empirical Student-$t$ calibration is not established here.

### 3.3 Budgets and comparators

We study fixed replication counts $n \in \{6, 7, 12, 24, 48, 96, 192\}$, the original adaptive $6 \to 12$ gate, and the four-look $6 \to 12 \to 24 \to 48$ gate. A reference decision $a_{192}(s, \xi)$ is computed with 192 replications on the same seed family $\xi$. Its purpose is comparison, not identification of the true optimal action. Reference stability is separately measured across independent seed families.

For an adaptive rule $A$ with empirical mean cost $\bar{B}_A$, an expected-budget-matched randomized comparator chooses a high fixed budget $n_H$ with probability

$$p_A = \frac{\bar{B}_A - n_L}{n_H - n_L}, \qquad n_L \le \bar{B}_A \le n_H. \tag{4}$$

The original adaptive rule is matched against a 6/12 mixture; the four-look rule against a 6/48 mixture. The expected agreement of this state-independent mixture is $(1 - p_A)G_{n_L} + p_A G_{n_H}$. The mixing probability is computed after observing the adaptive mean budget. Matching is therefore retrospective and holds only for expected replication counts under the observed state distribution; it is not a prospectively randomized, equal-realized-cost experiment. It does not establish dominance over all feasible budget policies.

## 4 What the measured regret represents

### 4.1 Reference agreement and score regret

For a policy $A$ evaluated on seed $j$ and frozen state $s$, reference agreement is

$$G_A = \frac{1}{JS}\sum_{j=1}^{J}\sum_{s=1}^{S}\mathbf{1}\{a_A(s, j) = a_{192}(s, j)\}, \qquad J = 30,\ S = 66. \tag{5}$$

This treats the $JS = 1980$ seed–state pairs as descriptive observations; the 30 seeds do not create 30 independent market histories.

An independent evaluation stream of 24 scenario batches produces an action score estimate $\widetilde{U}_j(a; s)$ for each training-frozen candidate. Let $\widetilde{\mathcal{A}}(s, j)$ denote candidates with an available independent score. The reported simulation-score regret proxy is

$$\widetilde{R}_A(s, j) = \max_{b \in \widetilde{\mathcal{A}}(s,j)} \widetilde{U}_j(b; s) - \widetilde{U}_j(a_A(s, j); s), \tag{6}$$

when the selected action has a score. The numerical implementation truncates at zero to guard against floating-point or menu inconsistencies. The feasible-score version maximizes over candidates passing all 24 evaluation feasibility checks and is reported only when the selected action is also evaluation-feasible. Missing scores are never set to zero. The all-method score common support is only 965/1,980 (48.74%), and the all-method evaluation-feasible support is 954/1,980 (48.18%); both are selected subsets. This proxy is expressed in engine score units, not dollar returns or realized portfolio utility. Its sample maximum can be optimistic, and the selected action may be absent from the scored menu after a fallback. Differences of order $10^{-6}$ cannot be translated into portfolio dollars, turnover, or investor welfare without a validated mapping from the engine

score and its normalizer to economic outcomes. Statistical visibility of the contrast is not evidence of economic materiality.

For comparison with a reference action we also calculate the signed gap $\widetilde{U}_j(a_{192}; s) - \widetilde{U}_j(a_A; s)$ where both scores exist. The signed gap can be negative: the finite-budget reference is not an oracle. The paper does not call this quantity implementable-policy regret or hindsight trading regret. Those concepts require a separate policy class, filtration, and market outcome law.

## 4.2 A finite-action bound and its limits

The following standard inequality explains why uniform score accuracy matters for action selection, without making an unsupported claim about the observed estimator.

**Proposition 1** (Uniform score error)**.** *For a fixed nonempty finite action set $\mathcal{A}_0$, let $a^* \in \arg\max_{a\in\mathcal{A}_0} U(a)$ and $\widehat{a} \in \arg\max_{a\in\mathcal{A}_0} \widehat{U}(a)$. If $\max_{a\in\mathcal{A}_0} |\widehat{U}(a) - U(a)| \le \varepsilon$, then $U(a^*) - U(\widehat{a}) \le 2\varepsilon$.*

*Proof.* Insert $\widehat{U}(a^*)$ and $\widehat{U}(\widehat{a})$ between $U(a^*)$ and $U(\widehat{a})$. The middle difference is nonpositive by the definition of $\widehat{a}$, while each remaining term is at most $\varepsilon$. □

The statement applies to a common action set and a uniform error bound. It does not imply that the Student-$t$ gate has such a bound, or that (6) estimates the unknown optimal market action's regret. In particular, $\mathcal{A}_n(s)$ in (1) varies with the budget, violating the common-set premise unless feasibility is held fixed.

## 4.3 Why all-rep eligibility changes the estimand

**Proposition 2** (Nested eligibility)**.** *For fixed candidate identities and a common nested sequence of feasibility indicators $F_1, F_2, \ldots$, the eligibility sets defined by (1) satisfy $\mathcal{A}_{n+1}(s) \subseteq \mathcal{A}_n(s)$ for every $n$. If $F_r(a; s)$ are independent Bernoulli variables with common success probability $p_a$, then $\Pr(a \in \mathcal{A}_n(s)) = p_a^n$.*

*Proof.* If $a$ is eligible after $n+1$ checks, all first $n$ checks also equal one. Under the additional independence and common-probability assumptions, multiply the $n$ success probabilities. □

The nesting result is deterministic; the $p_a^n$ expression is an illustrative model, not an estimate of the experiment's feasibility law. Define instead $\mathcal{A}^{(m)}(s) = \{a \in \mathcal{A}_{\text{train}}(s) : \prod_{r=1}^{m} F_r(a; s) = 1\}$ with fixed $m \le n$. Holding $m = 6$ constant as scoring budget $n$ grows separates the eligibility sample from additional scoring replications. It does not certify population chance constraints, and it uses only a short fixed prefix to screen feasibility.

## 4.4 What equal-budget improvement establishes

A state-dependent stopping rule may allocate its extra replications to a subset of states. The next identity makes the relevant comparison precise and also shows why matching only mean budget is insufficient to establish optimality.

**Proposition 3** (Equal-budget mixture identity)**.** *Fix a probability distribution $\mu$ over seed–state pairs and a binary evaluation outcome $g_n(s, \xi) \in \{0, 1\}$ for each fixed replication budget $n$. Let $n_L < n_H$ and let $p \in [0, 1]$ be constant across states. A randomized rule that chooses $n_H$ with probability $p$ independently of $(s, \xi)$ has expected cost $n_L + p(n_H - n_L)$ and expected evaluation outcome $(1-p)\mathbb{E}_\mu[g_{n_L}] + p\mathbb{E}_\mu[g_{n_H}]$. If an adaptive rule $A$ has the same expected cost, its improvement over the mixture is*

$$\Delta_A = \mathbb{E}_\mu[g_A] - (1-p)\mathbb{E}_\mu[g_{n_L}] - p\mathbb{E}_\mu[g_{n_H}]. \tag{7}$$

*Proof.* The mixture's cost and evaluation outcome are conditional expectations given $(s, \xi)$. Average first over the independent Bernoulli budget draw and then over $\mu$; linearity of expectation gives both identities. □

This proposition is an accounting identity rather than a new statistical optimality theorem. In particular, $\Delta_A > 0$ may depend on the comparator's chosen budgets, the finite-reference metric $g$, and the empirical distribution $\mu$. It does not imply $A$ dominates every state-dependent allocation with the same expected cost. The present results evaluate (7) using the empirical distribution of 66 states and 30 seeds, with $p$ obtained from the realized mean cost.

### 4.5 Conditional finite-sample testing statement

It is possible to justify a multiple-look familywise bound, but only if the individual comparisons are themselves calibrated. The distinction is material because our implementation uses a Student-$t$ approximation.

**Proposition 4** (Union-bound error control under valid marginal tests)**.** *Suppose at most $K$ prespecified checkpoints and $M$ prespecified one-sided null hypotheses are tested. If, for every true null and each checkpoint, the corresponding test rejects with probability at most $\alpha/(KM)$ under the joint data-generating law, then the probability of one or more false rejections is at most $\alpha$. No independence between checkpoints or comparisons is required.*

*Proof.* The event of any false rejection is contained in the union of at most $KM$ false-rejection events. Sum their marginal probability bounds. □

The proposition does not establish that the tests used here satisfy its premise. Scores can be non-Gaussian, share simulated market scenarios, and be filtered through a feasibility procedure. A Student-$t$ critical value has an exact finite-sample interpretation under stronger distributional assumptions than those verified in this application. Moreover, early default and challenger selection depend on several comparisons. Equation (3) is therefore described as a nominal, multiplicity-adjusted gate; it is not advertised as an anytime-valid confidence sequence or a certified $\alpha$-safe trading rule.

### 4.6 Algorithm and computational unit

**Algorithm 1 (four-look holdout gate, implemented logic).**

1. Given a frozen state, run a 300-scenario training evaluation; retain the highest-scoring training-feasible candidate for each action type. If no IYW default survives training, return an infeasible fallback.
2. For $r = 1, \ldots, 48$, evaluate the frozen candidates using a 250-scenario holdout batch with a predetermined seed. Store paired candidate scores and feasibility indicators.
3. At $r \in \{6, 12, 24, 48\}$, form the eligible menu by requiring feasibility in every accumulated batch. If IYW is ineligible, return an infeasible fallback.
4. Compute paired Student-$t$ bounds using (3). Accept a unique challenger only when its lower bounds exceed the default and all other eligible candidates by more than $\delta$.
5. At $r < 48$, stop with the IYW default if all challengers have upper bounds below $-\delta$, or if IYW is the only eligible candidate. Otherwise continue.
6. At $r = 48$, return IYW if no challenger passed. Record the decision identity, replication count, and stopping reason.

One replication is a complete re-evaluation of the candidate menu under 250 simulated scenarios, not one simulated daily return or one scalar score draw. Mean replication count is therefore a proxy for computational expense; training simulations, portfolio-state reconstruction, and software overhead are not included in the headline budget metric. If evaluation cost varies materially with state or action menu size, wall-clock or operation-weighted comparisons would be required before making claims about actual runtime savings.

## 5 Data, implementation, and experimental design

### 5.1 Frozen market trajectory

The underlying historical backtest uses retrospectively adjusted daily closing prices downloaded through Yahoo Finance for MSFT, AAPL, AMZN, GOOGL, NVDA, AMD, and IYW. The stored price window begins in January 2015 and ends in December 2024; the evaluation starts on 2 January 2018. The frozen snapshot is identified by SHA-256 `258a89e87ad0878276c91ddbf3d31f20d0084862d57cab4472d04b80fd41f093`. The equity universe was chosen retrospectively and contains surviving firms; it is not a point-in-time historical universe. Adjusted closing prices and a next-trading-day execution approximation are used. All policies receive the same scheduled cash contributions. The underlying engine was checked against an independent holdings/cash ledger, and its unitized NAV treatment was regression-tested after correcting the timing of transaction-fee recognition.

The primary budget experiments *do not* rerun 30 independent trading histories. They hold 66 decision states from the historical fixed-12 path constant, including holdings and cash, and vary the validation Monte Carlo stream. This controlled design isolates the gate at those states, but cannot identify the effect of replacing the gate on subsequent portfolio states. The historical terminal wealth comparison is therefore contextual, not the main identification design; its figures and accounting discussion are in Appendix A.

Table 1: Mean replication cost and agreement with fixed-192 reference, 30 holdout seeds and 66 frozen states. The fixed-192 row is self-agreement, not accuracy.

| Method | Mean replications | Agreement (%) | Disagreement (%) |
|---|---|---|---|
| Fixed 6 | 6.00 | 71.92 | 28.08 |
| Fixed 7 | 7.00 | 73.54 | 26.46 |
| Fixed 12 | 12.00 | 78.38 | 21.62 |
| Fixed 24 | 24.00 | 84.85 | 15.15 |
| Fixed 48 | 48.00 | 89.70 | 10.30 |
| Fixed 96 | 96.00 | 94.49 | 5.51 |
| Fixed 192 | 192.00 | 100.00 | 0.00 |
| Adaptive 6–12 | 6.94 | 76.41 | 23.59 |
| **Adaptive 6–12–24–48** | **9.64** | **82.73** | **17.27** |

Table 2: State-independent randomized mixtures matched on the adaptive rule's *observed mean* replication cost. Improvements are percentage points of agreement.

| Adaptive rule | Reference | Mean cost | Mixture (%) | Adaptive (%) | Gain (pp) |
|---|---|---|---|---|---|
| 6–12 | Fixed 96 | 6.94 | 77.85 | 81.41 | 3.56 |
| 6–12 | Fixed 192 | 6.94 | 72.93 | 76.41 | 3.49 |
| 6–12–24–48 | Fixed 96 | 9.64 | 78.38 | 87.88 | 9.50 |
| **6–12–24–48** | **Fixed 192** | **9.64** | **73.46** | **82.73** | **9.27** |

### 5.2 Seed allocation and diagnostics

A first 30-seed audit examined reference stability using independent holdout streams. A subsequent 30-seed batch, with separate seed offsets, compared the four-look method with fixed-budget gates and budget-matched mixtures. The 24-batch independent score evaluation used another seed family. The final feasibility diagnostic used three additional seeds and 66 states, yielding 198 paired seed–state observations per budget. The four-look design was motivated by preceding exploratory diagnostics; the later seeds are distinct simulation streams but the experiment as a whole is not preregistered.

The evaluation code records action identity, selected review horizon, allocation weights, replication count, fallback reason, and score coverage. Source code and prior results were left unchanged by the add-on audits. Seed-level descriptive intervals summarize variation over simulation seeds conditional on this fixed historical state panel. Because states share a market trajectory and the method was developed adaptively, these intervals are not confidence intervals for future market performance.

## 6 Results

Reference agreement is available for all 1,980 seed–state pairs, including fallback. Independent score-regret comparisons are conditional on common observed score support, which covers 965 pairs (48.74%). The progressively restrictive all-replication eligibility rule contributes to fallback actions with no comparable evaluation score. These two endpoints therefore describe different subsets and should not be read as estimates of the same population quantity.

### 6.1 Budget and reference agreement

The four-look method uses 9.64 replications per state on average and agrees with fixed 192 in 82.73% of comparisons (Table 1). Fixed 12 uses 12 replications and agrees in 78.38%. This direct comparison is informative, but its test thresholds and eligibility paths differ; it does not isolate the causal effect of adaptive stopping. Figure 1 displays the observed budget–agreement frontier for the tested rules, not a theoretical optimal frontier.

For the principal comparison the matched 6/48 mixture uses the 48-replication arm with probability 0.08658. Its expected agreement with fixed 192 is 73.46%, versus 82.73% for the adaptive rule, a 9.27-percentage-point difference. Across the 30 seed-level paired contrasts the descriptive interval is [8.35, 10.18] percentage points. The result concerns a state-independent randomized mixture and does not imply that the gate is optimal among adaptive allocations. The earlier two-look method shows a smaller but positive matched-budget difference.

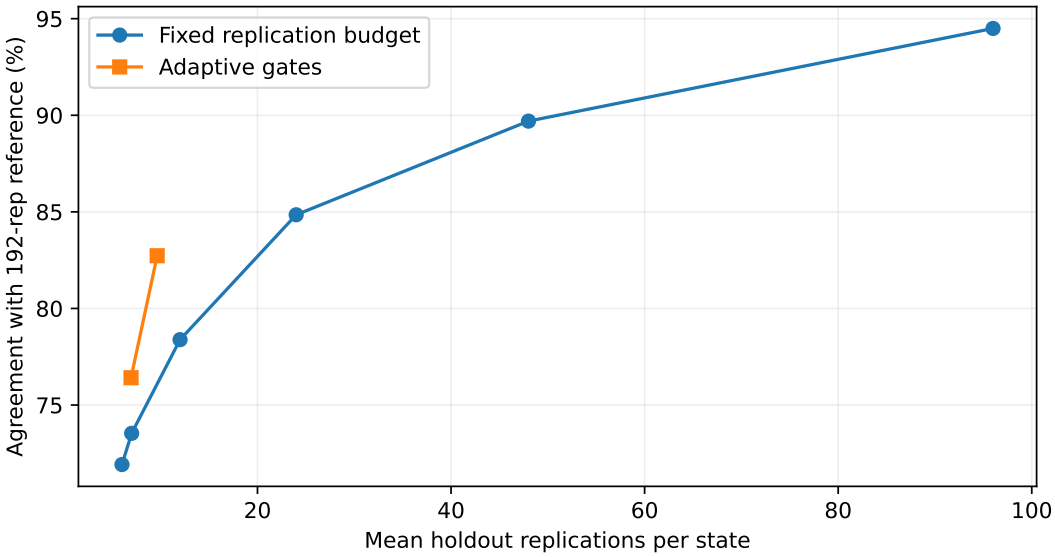



Figure 1: Reference agreement versus mean simulation budget. The reference is a finite-budget gate and is not an observed optimal action. Points share 66 historical states and 30 Monte Carlo streams.

Table 3: Paired adaptive-minus-budget-matched-mixture differences in independent simulation-score regret, conditional on common observed support. Negative values favor the adaptive rule. Units are engine candidate-score units, not investment returns. Intervals are descriptive across seeds.

| Adaptive | Support definition | $n$ | Difference | Descriptive interval |
|---|---|---|---|---|
| 6–12 | Three-method score | 1,343 | $-8.25 \times 10^{-7}$ | $[-1.15, -0.50] \times 10^{-6}$ |
| 6–12 | All-method score | 965 | $-9.79 \times 10^{-7}$ | $[-1.43, -0.53] \times 10^{-6}$ |
| 6–12–24–48 | Three-method score | 1,151 | $-1.85 \times 10^{-6}$ | $[-2.39, -1.31] \times 10^{-6}$ |
| **6–12–24–48** | **All-method score** | **965** | $\mathbf{-1.80 \times 10^{-6}}$ | $[-2.36, -1.25] \times 10^{-6}$ |

## 6.2 Reference variability across seeds

A separate 30-seed audit found that fixed-96 and fixed-192 decisions agree in 95.10% of within-seed comparisons. Fixed-192 decisions across *different* seed streams agree pairwise in 90.71% of comparisons; only 46 of 66 states had one modal decision selected in all 30 streams. These figures demonstrate residual reference variability. The fixed-192 comparator should not be labeled a ground-truth optimal policy, even if its replication count is substantially higher than the adaptive method's budget.

## 6.3 Independent score regret and common support

Independent evaluation scores are not observed for every chosen action. Out of 1,980 seed–states, 965 (48.74%) have a score for every method, and 954 (48.18%) have all methods' feasible-score-regret values. On the all-method score support, the four-look method's mean regret is $1.0390 \times 10^{-7}$, compared with $1.9190 \times 10^{-6}$ for the budget-matched mixture; the difference is $-1.8042 \times 10^{-6}$. On the larger three-method support its mean regret is $9.2507 \times 10^{-7}$. Table 3 reports the exact comparison contrasts to avoid drawing conclusions from rounded individual means. The corresponding all-method feasible-score contrast is $-1.8307 \times 10^{-6}$, with descriptive interval $[-2.3964, -1.2651] \times 10^{-6}$.

The support restriction matters. Score coverage falls from 73.13% for fixed 6 to 48.74% for fixed 192. The differences in Table 3 cannot be generalized to the 1,015 excluded observations. The observed improvement is measured in normalized engine-score units; its monetary and welfare relevance has not been established. Moreover, the score oracle in (6) is a maximum over noisy estimates. Thus the evidence concerns conditional simulation scores, not realized trading regret or a verified optimal policy. The signed reference-score gap is allowed to be negative, consistent with a noisy finite-budget reference.

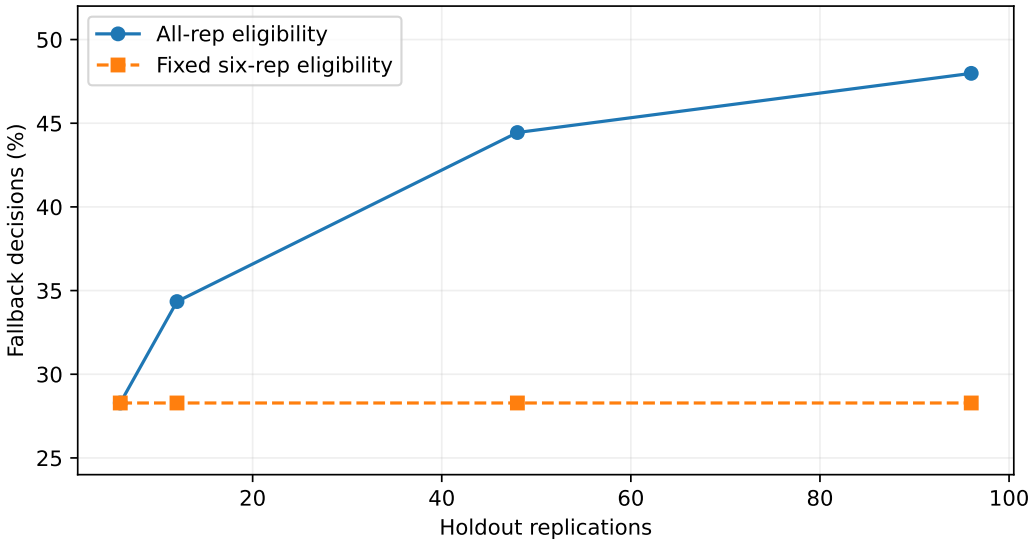



Figure 2: Fallback rates under all-replication versus fixed-prefix-six eligibility, based on three seed streams and 66 states. The latter fixes the screening sample as scoring replications increase; it is a diagnostic, not a validated chance constraint.

Table 4: Paired feasibility-rule diagnostic, 198 seed–state observations per budget. All changed decisions were all-replication fallbacks rescued by fixed-prefix-six screening.

| Budget | All-rep fallback (%) | Prefix-6 fallback (%) | Changed / 198 | Rescued |
|---|---|---|---|---|
| 6 | 28.28 | 28.28 | 0 | 0 |
| 12 | 34.34 | 28.28 | 12 | 12 |
| 48 | 44.44 | 28.28 | 32 | 32 |
| 96 | 47.98 | 28.28 | 39 | 39 |

## 6.4 Feasibility-screening diagnostic

The all-rep fallback rate increases monotonically from 28.28% to 47.98% as the budget rises from 6 to 96 (Table 4, Figure 2). Fixing the eligibility prefix at six keeps the fallback rate at 28.28% for every tested budget. At 96 replications, the two rules differ on 39 of 198 decisions, and each difference rescues a fallback under the original rule. This is a direct empirical consequence of coupling the feasible set to the number of simulated replications, consistent with Proposition 2. The diagnostic does not compare the resulting actions on realized market outcomes and does not establish that six feasibility checks are adequate for risk control.

## 6.5 Interpreting the severity of decision disagreements

Agreement is a zero–one measure. Two actions can disagree while having nearly identical estimated scores; conversely, a rare disagreement can carry a large score penalty. The independent evaluation therefore also reports disagreement-conditional score regret. In the 30-seed audit, the four-look method disagreed with fixed 192 on 342 seed–states, versus 204 for fixed 48. Its mean recorded score regret among disagreements was approximately $3 \times 10^{-6}$, versus approximately $4 \times 10^{-6}$ for fixed 48, subject to the methods' different score coverage. The medians were zero at the displayed precision. These conditional means are not a paired proof that the four-look gate prioritizes the most economically consequential states. The common-support contrasts in Table 3 provide the cleaner, albeit conditional, evidence.

## 6.6 Where the reported evidence ends

The principal outcome, agreement with fixed 192, is observable for every seed–state pair because the gate always returns a decision, including fallback. The score-regret outcome is missing for some fallback actions. As a result, the agreement and regret estimands have different supports. The feasibility diagnostic supplies an explanation for a large part of the missing-score pattern: the all-rep eligibility rule progressively rejects candidates as the number of checks increases. It does not recover the unobserved counterfactual score of an infeasible fallback.

Treating such missing values as zeros would change the estimand and spuriously favor methods with frequent fallback.

A second limitation concerns the high-budget reference. In the first independent seed experiment, fixed 96 and fixed 192 agreed on 95.10% of seed–state pairs, but fixed 192 had only 90.71% agreement across independent seed pairs. A reference decision can change because a candidate score lies close to a decision boundary or because eligibility is sensitive to a rare scenario. These explanations are consistent with the evidence but were not separated experimentally. It would be inappropriate to interpret $1 - G_A$ as a frequentist error rate against a true optimal action.

# 7 Discussion

The evidence is a paired audit of a particular validation implementation. Its checkpoints are fixed in advance within the evaluated algorithm, but the method was developed after exploratory diagnostics; the mixture cost weights were calibrated after observing the adaptive mean. These choices preclude treating the reported differences as preregistered confirmatory tests. The four-look rule concentrates additional replications on states that do not satisfy its stopping conditions. Relative to a randomized state-independent allocation at the same mean cost, it achieves greater finite-reference agreement and lower independently estimated score regret on common support. The same results leave important uncertainties. First, the high-budget reference changes across seeds; agreement is a reproducibility measure, not an error probability against an observed truth. Second, eligibility in the original implementation becomes more restrictive with larger budgets, which can increase fallback rates independently of score-estimation precision. Third, the four-look thresholds use Student-$t$ approximations whose coverage was not calibrated under the full candidate-selection and feasibility pipeline.

The state panel is taken from a single historical trajectory. Repeated Monte Carlo seeds quantify simulation variability conditional on that trajectory; they do not sample new market regimes, independent calendar periods, or alternative portfolios arising after different decisions. The universe is ex post and concentrated in large surviving technology firms. Retrospectively adjusted closes, scenario-model assumptions, execution approximations, and researcher-directed specification development limit any claim of investable alpha. The older manuscript's preference uncertainty and dynamic review-time optimization remain separate research questions; their untested conjectures are not used as theoretical support for the present gate results.

The nesting proposition isolates an eligibility mechanism, not a claim of statistical bias against a true chance-constraint target. Replacing the all-rep screen with a six-rep prefix is a diagnostic intervention, not a certified risk-control method. A practical implementation would define a feasibility target independently of computational budget and evaluate its statistical coverage separately. Fixed-prefix screening demonstrates the issue but is not a recommended replacement without risk validation. A stronger prospective evaluation would lock the method, candidate universe, and screening rule before observing a new market period. Those steps fall outside the present study.

# 8 Conclusion

Across 30 independent Monte Carlo holdout streams applied to 66 frozen historical portfolio states, a four-look validation gate used an average of 9.64 replications and achieved 82.73% agreement with a 192-replication reference. The matched-budget randomized comparator achieved 73.46%. Independent score comparisons favored the gate on the observable common support. A separate paired experiment established that requiring feasibility in every replication systematically increases fallback frequency with the replication budget. These findings identify a computational allocation benefit in the tested simulation environment and a specific mechanism that limits interpretation of high-budget references. They do not establish a portfolio-return advantage or finite-sample optimality.

# Declarations

**Data and code availability.** Research scripts and selected experimental outputs are available in the public GitHub repository: `https://github.com/SequentialDecisionResearch/ADTPA-Regret` (public repository; historical market snapshot not redistributed). The frozen historical market snapshot is not redistributed; full reruns require reconstructing the input data and checking data-version consistency. Market prices originate from Yahoo Finance via `yfinance`, subject to source terms.

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# A Historical portfolio context and accounting checks

## A.1 Historical portfolio context (descriptive only)

On the single frozen 2018–2024 historical path, the IYW benchmark and gate strategies had matched contributions. The fixed-12 and original adaptive-6-to-12 terminal wealths were approximately USD 1.505 million, against approximately USD 0.497 million for the IYW account. Their unit-NAV maximum drawdowns were approximately −54.32%, while the raw wealth and NAV drawdowns differ because of external contributions. The historical difference cannot be attributed to adaptive holdout validation alone; the stock universe, accumulation policy, and market path are major confounders. No full-path terminal-wealth estimate is reported for the four-look gate, which was evaluated at frozen states rather than deployed as a new sequential portfolio policy.

## A.2 An audit of the historical accounting comparator

The frozen historical experiment is relevant mainly as a source of portfolio states. The original IYW account follows the same dated external contribution schedule as the active policies, and an independent ledger reconstructs holdings, cash, purchases, and fees. The validation also checks that simulated scenarios generated at a decision date are invariant to removing all future rows of the observed price file. This is a useful no-future-access test of the scenario interface, but it does not establish that the retrospectively selected equity universe is point-in-time unbiased.

Table 5: Contextual historical backtest, one 2018–2024 market path. USD terminal wealth; NAV drawdown accounts for contributions. These are not results for the four-look gate.

| Policy | Terminal wealth (USD) | Unit-NAV max drawdown (%) | Holdout reps |
|---|---|---|---|
| IYW reference account | 497,330 | — | 0 |
| Fixed 6 gate | 1,506,843 | −54.37 | 402 |
| Fixed 12 gate | 1,504,662 | −54.32 | 792 |
| Adaptive 6–12 gate | 1,504,770 | −54.32 | 462 |

The backtest does not identify the gate's marginal contribution to portfolio returns. For example, fixed 6 has slightly higher terminal wealth than fixed 12 on this path despite lower agreement with the high-budget reference. There is no monotonic implication from computational reference agreement to realized terminal wealth. The absence of a sell-enabled comparator, a point-in-time equity universe, and an untouched prospective market period precludes an investable outperformance claim. The table is retained to disclose the origin of the 66 states and the scale of the underlying portfolio process, rather than as the principal evidence for the gate.

# B Reproducibility and result provenance

The attached project archive contains the frozen market snapshot and the additive audit scripts. The main figures and tables are derived from the following output files (relative to `ADTPA_IYW_Regret_Experiments_V3/`):

- `budget_matched_multistage_outputs`: reference agreement, budget mixtures, and descriptive intervals (Tables 1–2).
- `multi_seed_stability_outputs`: independent-seed reference stability.
- `decision_regret_budget_frontier_outputs`: independent evaluation score records.
- `common_support_audit_outputs`: common-support regret and coverage (Table 3).
- `budget_invariant_feasibility_outputs`: paired eligibility comparison (Table 4).
- `real_market_ablation_outputs`: contextual historical results and accounting checks.

The corresponding additive Python scripts are included in the supplied V3 research-code archive. Each accepts `-run` and produces CSV summaries and seed–state records. The present LaTeX archive contains the manuscript and plotting outputs, not a duplicate of the research code archive. The original historical snapshot and results must accompany any public reproducibility release subject to the data provider's terms.

# C Technical distinctions relevant to inference

The 30 seed streams are Monte Carlo replications of one fixed state panel. A seed-level interval estimates variability under the chosen pseudo-random scenario-generation protocol, conditional on the market states and design. It does not quantify uncertainty across market periods. The 66 states share holdings and calendar history; treating 1,980 seed–state observations as independent market draws would understate uncertainty. The budget-matched mixture probabilities were computed after observing the mean adaptive budgets. The independent evaluation stream prevents exact reuse of selection simulations, but it does not make a maximum of estimated action scores an unbiased oracle. In addition, the all-method common support is only 48.74% of seed–state pairs, and inference about excluded fallbacks requires a different estimand.